



## Important Formulas & Points to Remember: Connecting the Dots...

### Table of Contents

1. Representative Values
2. Mean
3. Median (Middle Value)
4. Range
5. Dot Plots

## Representative Values

When we have many numbers, we use a **single number** to represent them.

These are called **representative values**.

The two used in this chapter:

1. **Mean (Average)**
2. **Median**

## Mean

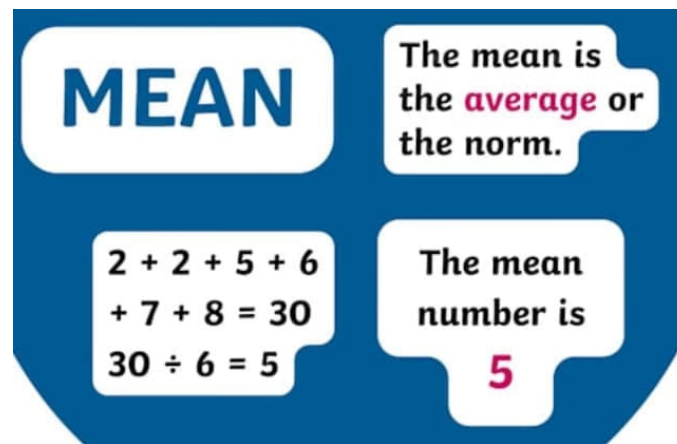
### Formula for Mean

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Mean is a **fair representation** of a group of numbers.

Mean acts like a **fair share**-if we redistribute values equally among all members, each gets

the **mean**.



## Median (Middle Value)

Median = the **middle value** when data is arranged in order.

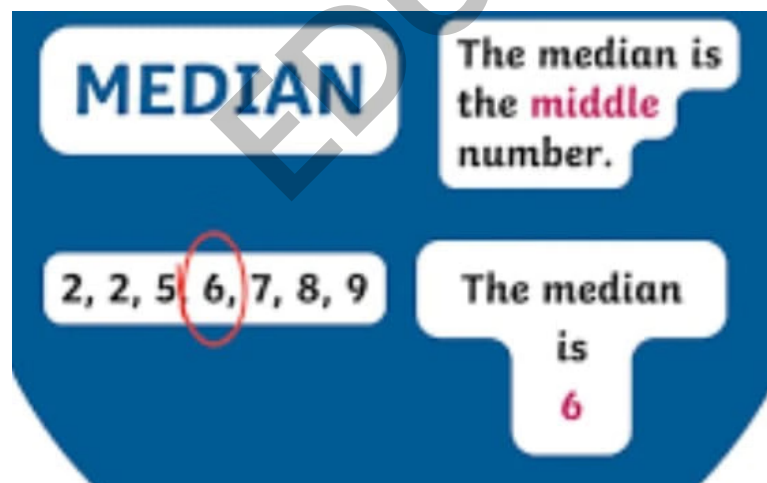
### Rules:

#### Case 1: Odd number of values

Median = middle term

#### Case 2: Even number of values

Median = average of the two middle values

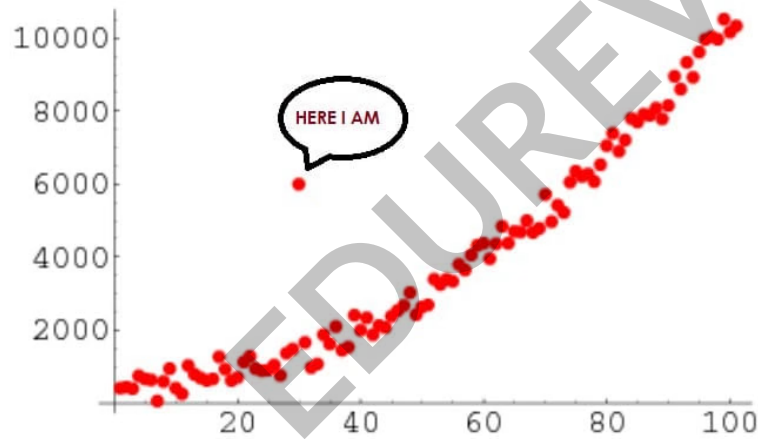


## Why Median?

Median is useful when data has **outliers** (extremely high or low values) that pull the mean too much.

## Outliers

An **outlier** is a value that is very **far** from the rest.



## Effect of Outliers:

Outliers **distort the mean**

But **median stays unaffected**

## When will mean and median differ?

A very small outlier  $\rightarrow$  Mean < Median

A very large outlier  $\rightarrow$  Mean > Median

This helps students interpret data properly.

## Range

Although not labelled separately, the chapter uses **range** during data comparison.

Range Range = Maximum value - Minimum value

Range shows **spread** of data.

Higher range = more variation.

## Dot Plots

Dot plot shows data values as **dots placed above a number line**.

Useful to identify:

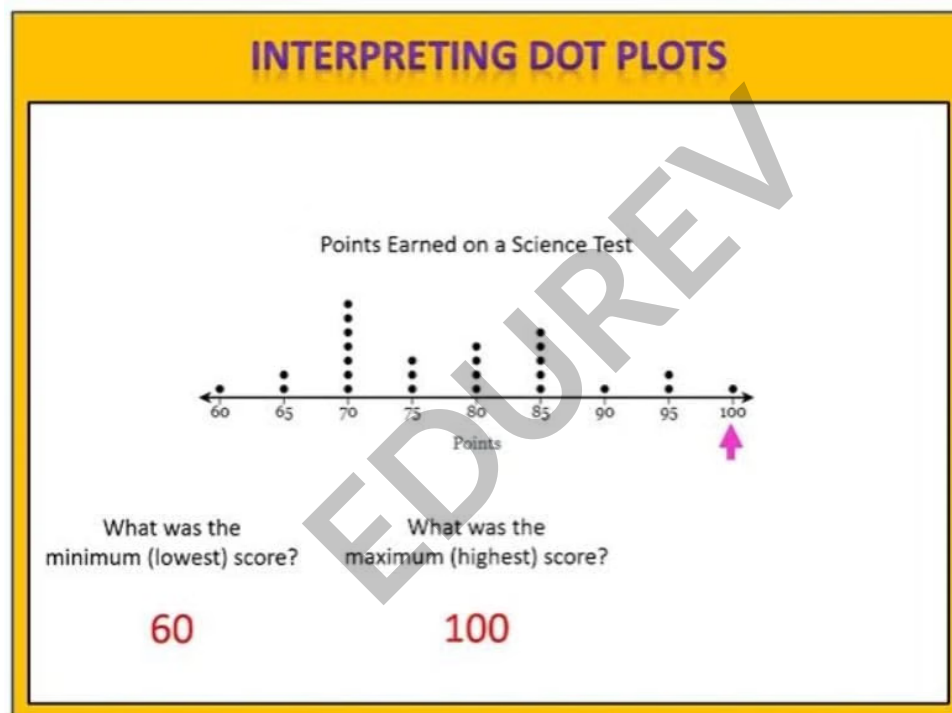
Clusters

Gaps

Outliers

Peaks (modes)

Range



## Interpreting Mean vs Median

If **mean > median**

→ There is a large value pulling mean upward.

If **mean < median**

→ A small value pulls mean downward.

If **mean ≈ median**

→ Data is fairly balanced.

## Zero vs. No Data

**0** means the participant participated but got zero.

**"-"** means the person **did not participate** in that case.

When calculating averages, divide only by **number of cases participated**, not total rows in the table.

## To Summarise:

| <b>Situation</b>          | <b>Best Measure</b>     | <b>Reason</b>             |
|---------------------------|-------------------------|---------------------------|
| Data has extreme values   | <b>Median</b>           | Not affected by outliers  |
| Values evenly spread      | <b>Mean</b>             | Represents fair balance   |
| Need to find consistency  | <b>Range</b>            | Small range → stable data |
| Need visual understanding | <b>Dot plot / Graph</b> | Shows spread & clusters   |